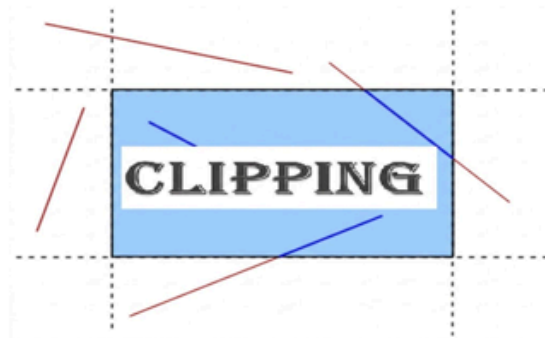


# Computer Graphics

## Line Clipping

### Lecture 11



## Clipping



- When we have to display a large portion of the picture, then not only scaling & translation are necessary, but also the visible part of the picture is identified. This process is not easy. Certain parts of the image are inside, while others are partially inside.
- For deciding the visible and invisible portion, a particular process called clipping is used. Clipping determines each element's visible and invisible portion. The visible portion is selected. An invisible portion is discarded.

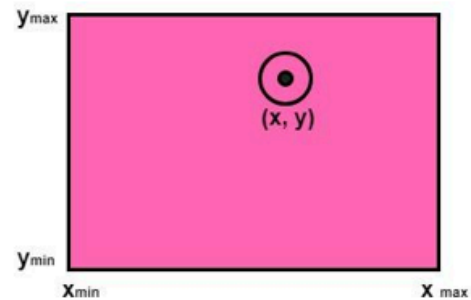
## Types of Clipping:



- Point Clipping
- Line Clipping
- Area Clipping (Polygon)
- Curve Clipping
- Text Clipping
- Exterior Clipping

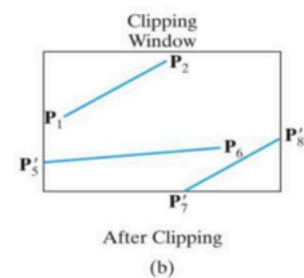


- iv.
- $y \geq y_{\min}$



- The concept of line clipping is the same as point clipping. In line clipping, we will cut the portion of the line that is outside of the window and keep only the portion that is inside the window.

- 



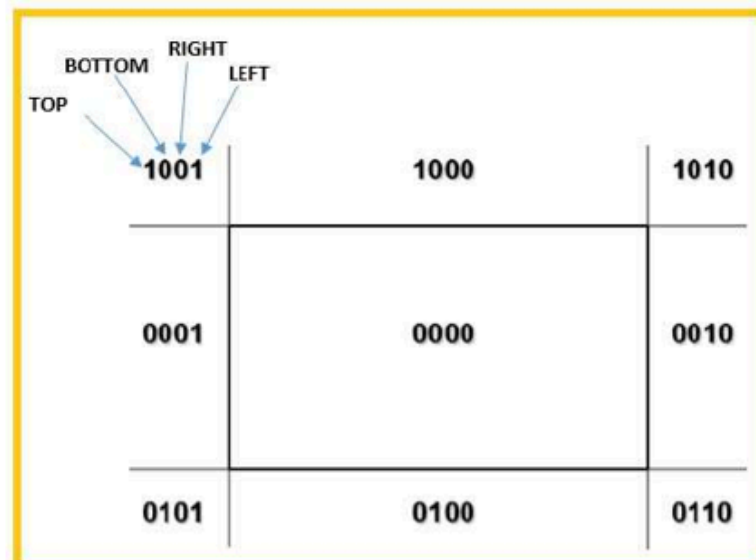
# Line clipping algorithms

The line clipping algorithms are:

- Cohen-Sutherland Line Clipping Algorithm
- Midpoint Subdivision Line Clipping Algorithm
- Liang-Barsky Line Clipping Algorithm

## Cohen-Sutherland Line Clipping Algorithm

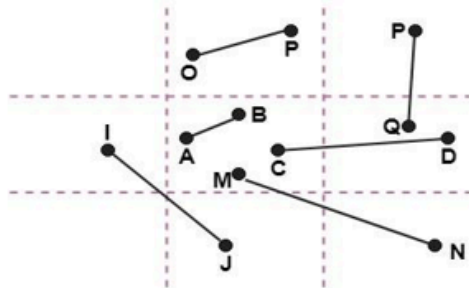
- Cohen-Sutherland Method [Widely used].
- In this method, region codes are used to identify the position of line endpoints relative to the rectangular clipping window.



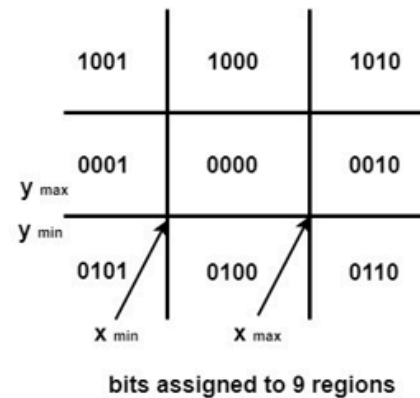
# Cohen-Sutherland Line Clipping Algorithm



**Problem 1:** The following figure show lines of various types

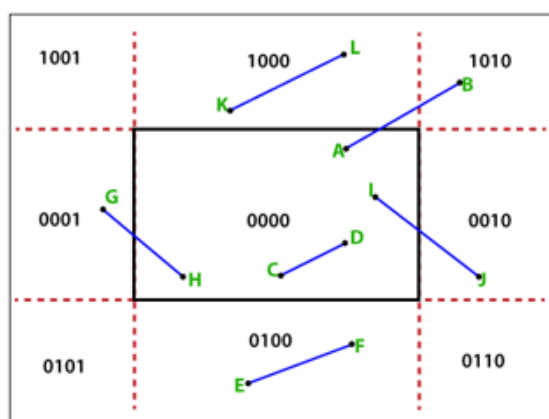


- Line AB is the visible case
- Line OP is an invisible case
- Line PQ is an invisible line
- Line IJ are clipping candidates
- Line MN are clipping candidate
- Line CD are clipping candidate



# Cohen-Sutherland Line Clipping Algorithm

**Problem 2:** In the below-mentioned example, we have different lines. The different category of the line—



- Line AB is a clipped case.
- The line CD is completely visible.
- Line EF is completely invisible.
- Line GH is a clipped case.
- Line KL is completely invisible.
- Line IJ is a clipped case.

# Cohen-Sutherland Line Clipping Algorithm



- **Step 1:** Calculate positions of both endpoints of the line
- **Step 2:** Perform OR operation on both of these end-points
- **Step 3:** If the OR operation gives 0000  
Then  
line is considered to be visible  
else  
Perform AND operation on both endpoints  
If And  $\neq$  0000  
then the line is invisible  
else  
And=0000  
Line is considered the clipped case.
- **Step 4:** If a line needs to be clipped, first find an intersection point of all boundaries with the following formula:  
$$m = (y_2 - y_1) (x_2 - x_1)$$

এক লাইনে মনে রাখো:

**OR = 0  $\rightarrow$  Visible**

**AND  $\neq$  0  $\rightarrow$  Invisible**

**Otherwise  $\rightarrow$  Clipping Candidate**

# Cohen-Sutherland Line Clipping Algorithm



- **Step 5:** Finding intersection points  
i) y value of vertical line :

When the line intersects the left-side boundary of the window port.

$$y = y_1 + m(x - x_1)$$

Here  $x = x_{\min}$  (Minimum value of x coordinate)

When the line intersects the right-side boundary of the window port.

$$y = y_1 + m(x - x_1)$$

Here  $x = x_{\max}$  (Maximum value of x coordinate)

- ii) x value of horizontal line:

When the line intersects Top side boundary of the window port.

$$x = x_1 + (y - y_1)/m$$

Here  $y = y_{\max}$  (Maximum value of y coordinate)

When the line intersects the bottom side boundary of the window port.

$$x = x_1 + (y - y_1)/m$$

Here  $y = y_{\min}$  (Minimum value of y coordinate)

## Point Outside at Left

$$x = x_{\min}$$

$$y = y_1 + (x_{\min} - x_1)m$$

## Point Outside at Right

$$x = x_{\max}$$

$$y = y_1 + (x_{\max} - x_1)m$$

## Point Outside at Bottom

$$x = x_1 + (y_{\min} - y_1)/m$$

$$y = y_{\min}$$

## Point Outside at Top

$$x = x_1 + (y_{\max} - y_1)/m$$

$$y = y_{\max}$$

## ✂ 4. TOP boundary এ clip করলে

যদি  $y = y_{\max}$

$$x = x_1 + (x_2 - x_1) \frac{y_{\max} - y_1}{y_2 - y_1}$$

নতুন point:

$$(x, y_{\max})$$

### 3. BOTTOM boundary এ clip করলে

যদি  $y = y_{min}$

$$x = x_1 + (x_2 - x_1) \frac{y_{min} - y_1}{y_2 - y_1}$$

নতুন point:

$$(x, y_{min})$$

### 2. RIGHT boundary এ clip করলে

যদি  $x = x_{max}$

$$y = y_1 + (y_2 - y_1) \frac{x_{max} - x_1}{x_2 - x_1}$$

নতুন point:

$$(x_{max}, y)$$

### 1. LEFT boundary এ clip করলে

যদি  $x = x_{min}$

তাহলে,

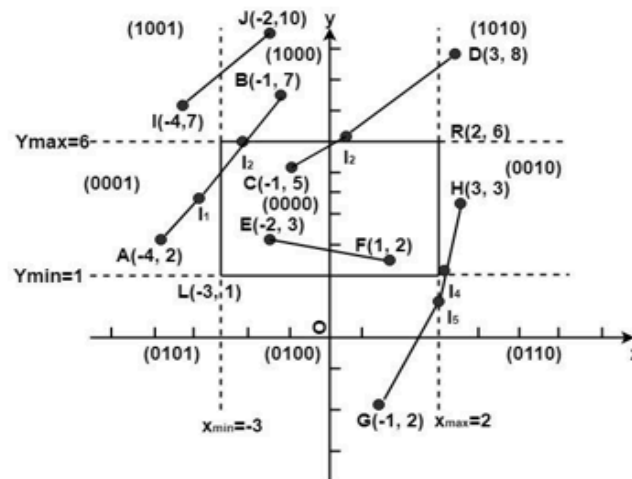
$$y = y_1 + (y_2 - y_1) \frac{x_{min} - x_1}{x_2 - x_1}$$

নতুন point:

$$(x_{min}, y)$$

## Example of Cohen-Sutherland Line Clipping Algorithm:

Let R be the rectangular window whose lower left-hand corner is at L (-3, 1) and whose upper right-hand corner is at R (2, 6). Find the region codes for the endpoints in fig.



The region code for point (x, y) is set according to the scheme

|                  |                   |
|------------------|-------------------|
| A (-4, 2) - 0101 | F (1, 2) - 0000   |
| B (-1, 7) - 1000 | G (-1, -2) - 0100 |
| C (-1, 5) - 0000 | H (3, 3) - 0010   |
| D (3, 8) - 1010  | I (-4, 7) - 1001  |
| E (-2, 3) - 0000 | J (-2, 10) - 1000 |

We place the line segments in their appropriate categories by testing the region codes found in the problem.

**Category 1 (visible):** EF since the region code for both endpoints is 0000.

**Category 2 (not visible):** IJ since (1001) AND (1000) = 1000 (which is not 0000).

**Category 3 (candidate for clipping):**

AB since,

(0001) AND (1000) = 0000,

CD since,

(0000) AND (1010) = 0000, and

GH. since (0100) AND (0010) = 0000.

The candidates for clipping are AB, CD, and GH.



### For Clipping AB,

A (-4, 2), B (-1, 7)  
 $x_1 = -4, y_1 = 2$   
 $x_2 = -1, y_2 = 7$   
 $m = (7-2) / (-1+4) = 5/3$

#### In case of A,

A is outside the window, and the line intersects the left-side boundary of the window port.  
Here,  $x_{min} = -3$ .

$$y = y_1 + m(x - x_1) = 2 + 5/3 (-3+4) = 11/3$$

The resulting intersection point is  $I_1 (-3, 11/3)$  or  $(-3, 3)$

#### In case of B,

B is outside the window (i.e., its code is 1000)  
the line that intersects the top-side boundary of the window port.  $y_{max} = 6$ .

$$x_0 = x_1 + (y - y_1) / m = -4 + (6-2) * 3 / 5 = -4 + 12/5 = -8/5 = -1$$

### For clipping CD,

C (-1, 5), D (3, 8)

$x_2 = -1, y_2 = 5$

$x_1 = 3, y_1 = 8$

$$m = (5-8) / (-1-3) = 3/4$$

we start with D since it is outside the window. Its code is 1010.

The line intersects Top side boundary of the window port.  $y_{max} = 6$ .

$$x = x_1 + (y - y_1) / m = 3 + (6-8) * 4/3 = 3 - 8/3 = 9-8/3 = 1/3$$

$y_{max} = 6$ . The resulting intersection  $I_3$  is  $(1/3, 6)$ , and its code is 0000.

### For clipping GH,

we can start with either G or H since both are outside the window.

G (1, -2), H (3, 3)

$$m = (3+2) / (3-1) = 5/2$$

In case of G,

the line intersects the bottom side boundary of the window port.

$$x_0 = x_1 + (y - y_1) / m = 1 + ((-1+2) * 2) / 5 = 1 + 6/5 = 11/5 = 2.2$$

The code for G is 0100 line  $y_{min} = 1$ .

The resulting intersection point is  $I_4 (2, 1)$  and its code is 0010. We clip  $GI_4$  and work on  $I_4H$ . Segment  $I_4H$  is not displaying since  $(0010) \text{ AND } (0010) = 0010$ .

### Example

ধরি original line:

$$A(x_1, y_1), \quad B(x_2, y_2)$$

Clipping করার পর:

$$A' = (x_{A'}, y_{A'}), \quad B' = (x_{B'}, y_{B'})$$

তাহলে final answer এ এভাবে লিখবে:

$$\boxed{\text{Clipped line segment} = A'B'}$$

অর্থাৎ,

$$\boxed{A' = (x_{A'}, y_{A'}), \quad B' = (x_{B'}, y_{B'})}$$

এবং শেষে:

$$\boxed{\therefore \text{Clipping part of } AB = A'B'}$$

ধরো হিসাব করে পেলে

$$A' = (2, 4.167)$$

$$B' = (8, 5.167)$$

তাহলে final answer:

Therefore, the clipped line segment is  $A'B'$ , where

$$\boxed{A' = (2, 4.167), \quad B' = (8, 5.167)}$$